

CardioTwin AI — System Architecture & Methodology

Digital Twin Challenge 2026 by Happiest Health • Technical Architecture Document

1. Executive Overview & Core Philosophy

CardioTwin AI creates a dynamic, personalized digital twin of high-risk cardiovascular patients (hypertension, type-2 diabetes, dyslipidemia) by combining static Electronic Health Record (EHR) profiles with continuous wearable sensor telemetry. Rather than treating health as a static snapshot, the system continuously calibrates a personalized physiological baseline, quantifies multivariate drift, forecasts multi-horizon deterioration risks (6h, 24h, 72h), and provides a deterministic What-If simulator for clinician decision support.

2. End-to-End System Pipeline

Layer	Module	Primary Functionality
Data Ingestion	Synthetic EHR & Correlated Wearable Stream	Correlated cardiovascular priors (HTN, T2D, Lipids) + Real-time telemetry
Data Quality	PPG Artifact & Plausibility Filter	Sensor disconnection, ectopic beat, and motion artifact screening
N=1 Baseline	Personal Baseline Engine	Empirical 95% CI (mean $\pm$ 2 SD) across quiescent calibration windows
Twin State	Multivariate Twin Drift Engine	Twin Drift Score (0-100) representing distance from personal baseline
Forecasting	Multi-Horizon Risk Forecaster	Calibrated 6h, 24h, and 72h deterioration probabilities
Explainability	SHAP / Contribution Decomposition	Exact mathematical weight attribution per baseline deviation
Simulation	What-If Counterfactual Simulator	Clinician-configurable behavioral & therapeutic response trajectories
Interface	Clinician Command Dashboard	React/TypeScript dashboard with timeline progression & ABDM/FHIR layer

3. Multimodal Ablation Study Results

Model Architecture	Input Modalities	AUROC	AUPRC	F1	Lead Time
Model A (EHR-Only)	Age, BMI, BP, HbA1c, Meds, Med Adherence	0.487	0.209	0.161	2.1 hours
Model B (Wearable-Only)	Raw Wearable Telemetry + Rolling Dynamics	0.648	0.380	0.388	7.2 hours
Model C (Digital Twin Fusion)	EHR + Wearables + Baseline Deviations + Drift	0.715	0.482	0.431	14.5 hours

4. India-Scale ABDM Readiness & Privacy Invariants

- **HL7 FHIR R4 Ready:** Structures observations using LOINC 8867-4 and NRCES India profiles.
- **ABHA Integration Direction:** Designed to map into ABDM Health Information Provider (HIP) bundles.
- **Zero Real PII:** 100% synthetic mathematical data. Zero real-patient health records used.
- **Compliance Notice:** ABDM/FHIR compatibility is an architectural direction, not official government certification.